

***Technical
Writing***



Technical Writing

Lecture (2)

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Course Reference:

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Johnson-Sheehan, R., “Technical Communication Strategies for Today”, 2nd ed., Pearson, 2014.

Definition/Purpose of Course?

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- Technical communication is the “**art** and **science** of making complex technical information **accessible**, **usable**, and **relevant** to a variety of audiences in a variety of settings.”
- **Composition** of the types of documents technical professionals write.
- **Review** of the planning, drafting, and editing processes.
- **Interaction** with new technologies and new environments..

Accessibility

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- **Accuracy**— no errors of fact or grammar.
- **Clarity**— no ambiguity.
- **Completeness**— all necessary information is present.
- **Diction**—appropriate and grammatically correct language.
- **Organization**—logical arrangement of parts.
- **Visual effectiveness**— page/screen design, graphics..

Usability

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- **Allows** audience to **perform** the task or **retrieve** the information they need.
- **Studies** design of table of contents, index, headings, etc.
- **Keeps** learning curve short..

Relevance

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- **Focus** on your audience's need for information.
- **Give** only what's needed.
- **Use** language that fits your audience and situation and is consistent..

On-the-job Writing

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- **Communicate expertise** to clients, customers, and the public.
- **Report** tech activities to supervisors and others.
- **Write** proposals.
- **Instruct** lay people.
- **Correspond** with colleagues and clients..

Tech Communicator's Skills

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- **Facility** with language.
- **Use** of critical thinking skills to **solve problems**.
- **Ability** to assess situations, **determine** what the most important issues and the subsidiary ones.
- **Ability** to organize a document that presents the information clearly.
- **Awareness** of the persuasive nature of writing.
- **Ability** to compose documents that convince your reader(s) to act as you suggest or to trust your point of view.
- **Ethical behavior** shown by avoiding **deceptive** graphics, presenting information truthfully,
- **Balancing** ethical and moral beliefs with the demands of the job..

Communicating in the Technical Workplace

What is Technical Writing?

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It is **written communication** for engineer, scientist, doctor, nurse, psychologist, social worker, anthropologist, architect, technical writer, or any other professional in a technical field to **deliver** information with the intent of achieving a desired purpose.



Genre and the Technical Writing Process

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Genres:

- Relatively **stable** patterns that reflect the activities and practices of the workplace.
- **Shape** a document's content, organization, style, design, and medium
- **Help** you anticipate the needs of your readers and their situations..

Genre and the Technical Writing Process

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Genres:

- Analytical reports **follow** a different genre than technical specifications.
- They are **written** for different kinds of readers for different workplace situations.
- They **include** different kinds of information and follow their own organizational patterns.
- The style and design of these two genres are distinctly **different**..

Analytical Report	Specification
Introduction	Introduction
Methodology	Materials, List of Parts, Conditions, Tools Needed
Results	Step One
Discussion	Step Two
Conclusions/ Recommendations	.
Back Matter	Conclusion
	Troubleshooting (if needed)

Genre and the Technical Writing Process

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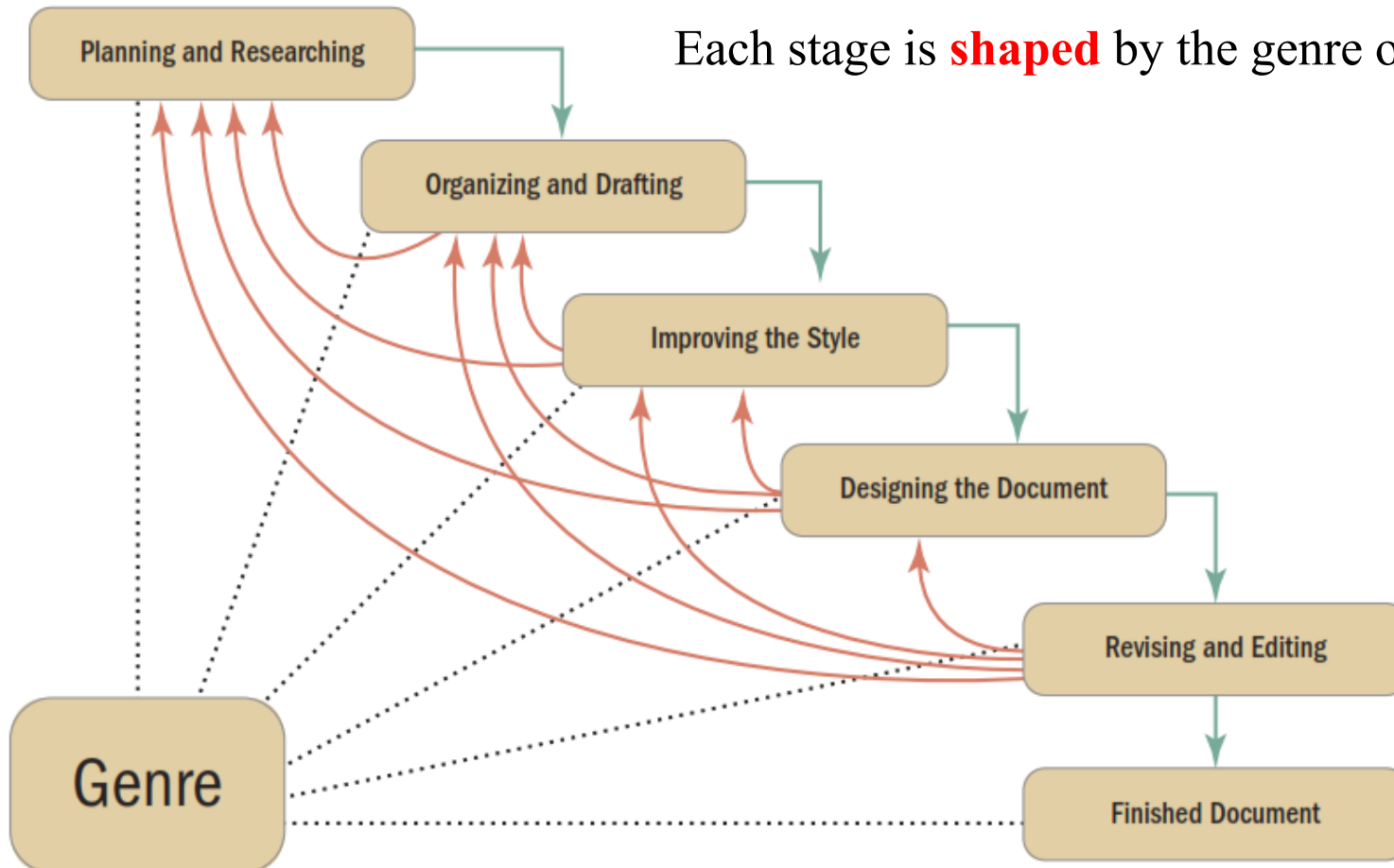
Technical writing is a process that **includes** the stages such as:

- 1- **Planning and researching:** planning the project, using research to **collect** information, and developing your own ideas.
- 2- **Organizing and drafting:** deciding how to **arrange** your information and then **turning** those ideas into sentences, paragraphs, and sections.
- 3- **Improving the style:** Writing **clearly** and **persuasively** for your readers.
- 4- **Designing the document:** developing an appropriate page design that **improves** the usability and attractiveness of your document.
- 5- **Revising and editing:** Improving the **quality** of your work by revisioning, rewriting, and proofreading your writing..

Genre and the Technical Writing Process

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The Technical Writing Process



The technical writing process involves **moving** back and forth **among** several stages.

Each stage is **shaped** by the genre of the document.

Genre and the Technical Writing Process

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- While drafting, you may **discover** that you need to do more research on your topic.
- While editing, you may **decide** that you need to draft an additional section for the document.
- Overall, these stages will **lead** you from the beginning of a project to the end..

Stage 1: Planning and Researching

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When planning and researching, you should **spend** some time doing three activities:

- A- Define the rhetorical situation: **Identify** your document's subject, purpose, readers, and context of use.
- B- State your purpose: **Sharpen** your purpose into a one-sentence statement that will guide your research and drafting of the document.
- C- Research your subject: **Use** electronic, print, and empirical sources to collect information on your subject.

Stage 1: Planning and Researching

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A- Define the rhetorical situation:

The five **W** and **How** questions give you a sense of your rhetorical situation.

- **W**ho are my readers, and **w**ho else is involved with the project?
- **W**hat do the readers want and need, and **w**hat do I want and need?
- **W**here do they need the information, and **w**here will they use it?
- **W**hen will the information be used, and **w**hen is it needed?
- **W**hy do the readers need the information in this document?
- **H**ow should I achieve my purpose and goals?..

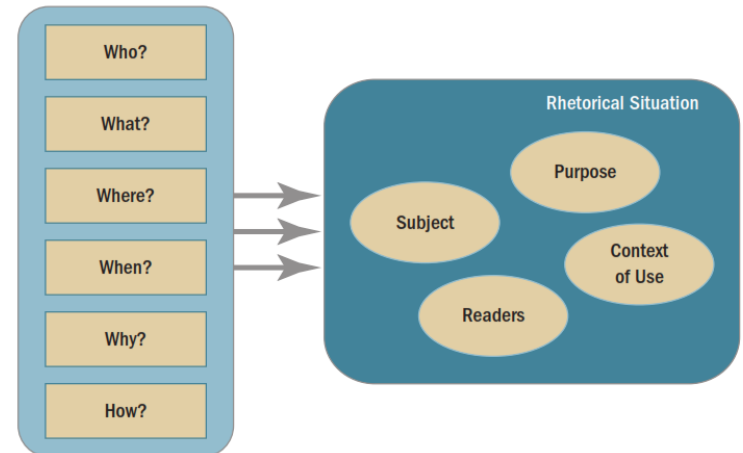
Stage 1: Planning and Researching

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A- Define the rhetorical situation:

Four elements of the rhetorical situation:

- **Subject** – what is document about? What is the scope of the project?
- **Purpose** – what is the document trying to achieve? Why do the readers need it? What do they need to know?
- **Readers** – who are the readers? What are their needs and interests? What are they looking for?
- **Context of Use** – where and when will the document be used? What constraints are there?



Stage 1: Planning and Researching

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B- State your purpose:

It is what you **want to do** and what you **want the document** to achieve.

- Your purpose statement is like a **compass** for the document. Once you have clearly **defined** your purpose for yourself and your readers, you can **use** that purpose statement to **guide** your decisions about the content, organization, style, and design of your document.
- When defining your purpose, try to **express exactly** what you want your document to achieve..

Stage 1: Planning and Researching

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B- State your purpose:

- Sometimes it helps to find an appropriate **action verb** and then build your purpose statement around it. Here are **some** useful action verbs that you might use:

Informative Documents

to inform
to describe
to define
to review
to notify
to instruct
to advise
to announce
to explain
to demonstrate

Persuasive Documents

to persuade
to convince
to influence
to recommend
to change
to advocate
to urge
to defend
to justify
to support

- The purpose of my report is to review
- The purpose of my proposal is to recommend

Stage 1: Planning and Researching

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C- Research your subject:

It is your **next step**.

- You need to **gather** information from a variety of sources, including the Internet, print documents, and empirical methods(e.g., experiments, surveys, observations, interviews).

Stage 2: Organizing and Drafting the Contents

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Two things are **done** at the same time:

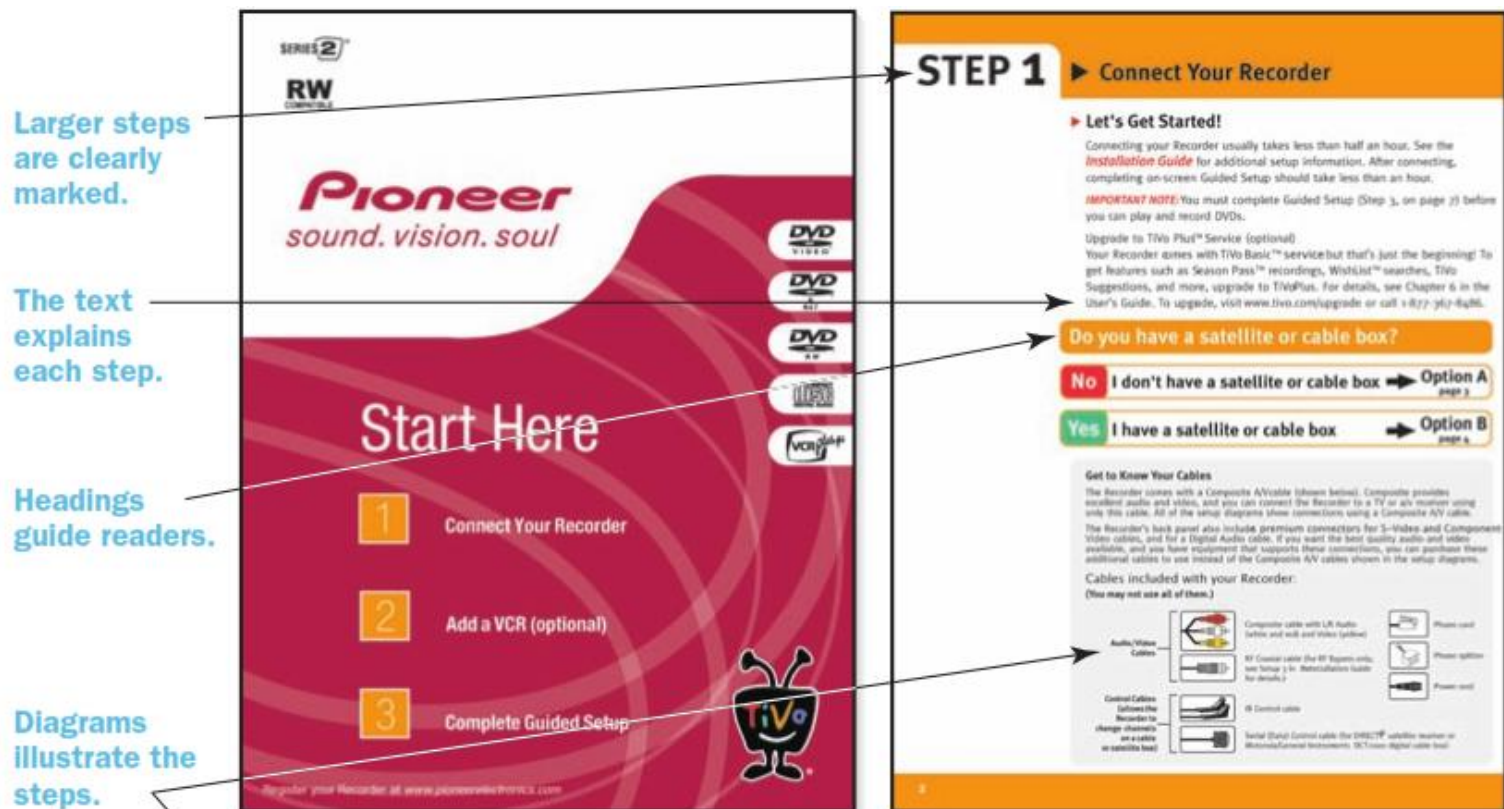
- 1- **Organizing the content:** Using **common genres** to shape your ideas into documents that will be familiar to readers.
- 2- **Drafting the content:** Generating the content of your document by **including** facts, data, reasoning, and examples.

If you **understand** the genre, you will **understand** how to organize the information you've collected in away that achieves your purpose.

Stage 2: Organizing and Drafting the Contents

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- A genre follows a pattern that readers will find **familiar**.
- Readers would **immediately recognize** this document as a set of instructions and be able to use it.



Stage 3: Improving the Style

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All documents **have a style, effective or not.**

Good style is a **choice** you **can** and **should** make.

Two kinds of style widely used in technical documents:

- 1- **Plain style:** This style **stresses** clarity and accuracy. By paying **attention** to your sentences and paragraphs, you can make your **ideas** clearer and easier to understand.
- 2- **Persuasive style:** You can use it to **motivate** readers by **appealing** to their values and emotions. You can **use** similes and analogies to add a **visual quality** to your work. You can use **metaphors** to change your readers' perspective on issues. Meanwhile, you can use **tone and pace** to add energy and color to your work..

Stage 4: Designing the Document

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- Your readers will expect your technical documents to be **well designed** with graphs, figures and charts.
- It can be **easily** designed by computer.
- As you **think** about the design of your document, keep this saying in mind: **Readers are “raiders”** for information.
- Your readers **want** the important parts **highlighted** for them.
- They **prefer** documents that use **effective** graphics and layout to make the information more accessible, interesting, and attractive.

Stage 4: Designing the Document

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Because readers are **raiders** of information, you want the design of your document to be **visually** accessible.

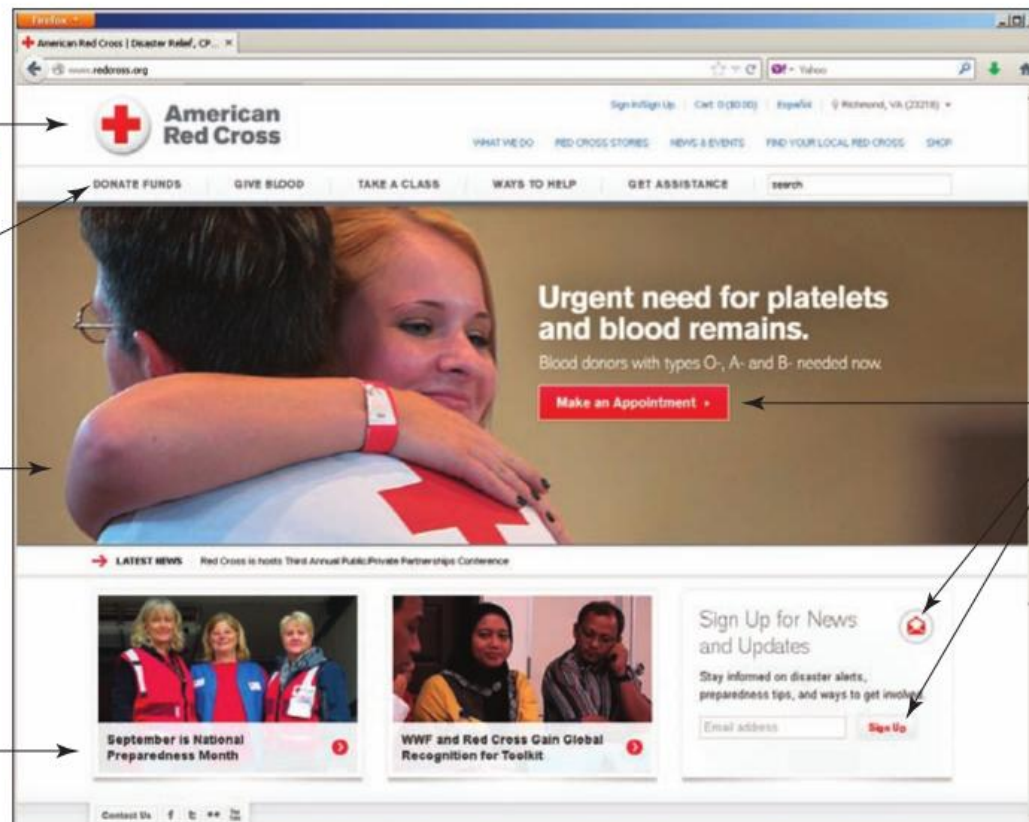
Document Design Is Very Important

The organization's name is easy to locate.

Topics are listed clearly.

Pictures add a human quality.

Text is highly scannable because it is in groups.



Color adds energy.

Stage 5: Revising and Editing

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- ❑ When you have **finished** drafting and designing the document, you are only a little over **halfway** finished.
- ❑ In technical communication, it is **crucial** to leave plenty of time for revising, editing, and proof reading.
- ❑ Clarity and accuracy are **essential** if your readers are going to understand what you are trying to tell them.
- ❑ There are **four** levels of revising and editing:
 - Level 1: Revising,
 - Level 2: Substantive editing.
 - Level 3: Copyediting.
 - Level 4: Proofreading..

Stage 5: Revising and Editing

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- level 1: Revising: Re-examine** your subject and purpose while **thinking** again, about the information your readers need to know.
- level 2: Substantive editing: Look** closely at the content, organization, and design of the document **to make sure** your readers can find the information they need.
- level 3: copyediting: Pay** close attention to the document's sentences, paragraphs, and graphics **to make sure** they are clear, accurate, and efficient.
- level 4: Proofreading:** Carefully **proofread** your document to eliminate grammar problems, typos, spelling errors, and usage mistakes..

Stage 5: Revising and Editing

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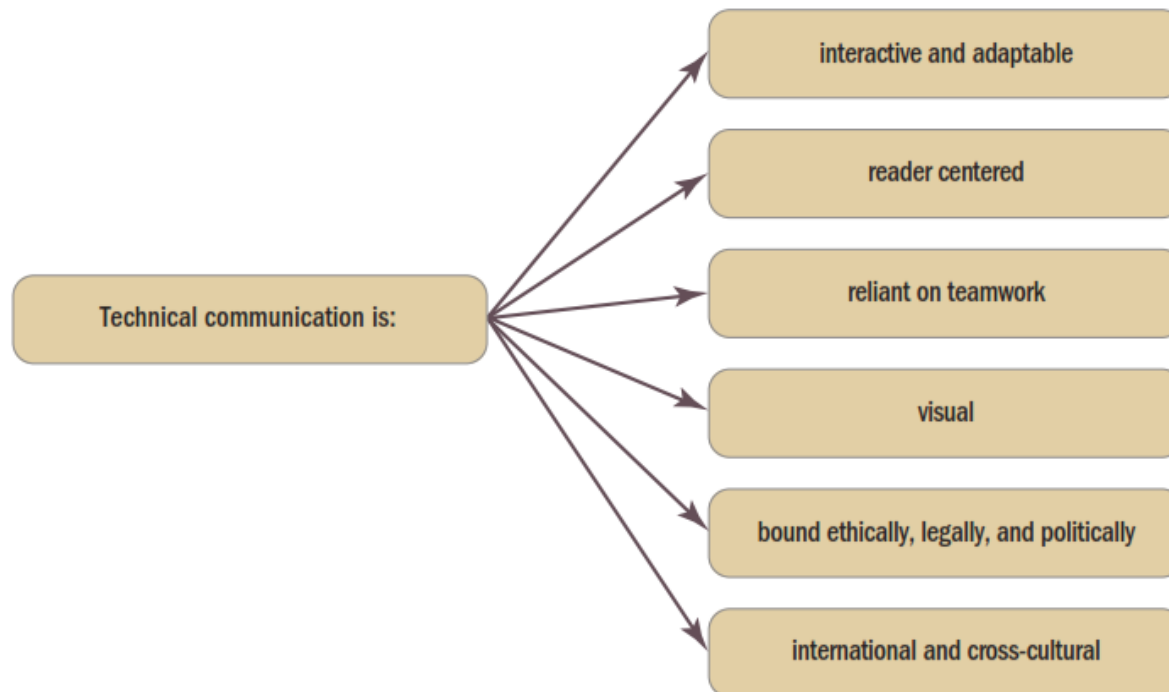
- In workplace documents, **errors** are a signal of low-quality work.
- Revising and editing is a **crucial step** in the technical workplace, where clarity and accuracy are **essential**.
- Your supervisors will ask you to do **much more** revising and editing than your college professors..

What Is Technical Communication?

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Technical Communication is a **process** of managing technical information in ways that allow people to **take** actions.

Technical communication puts much more **emphasis** on managing information and taking action **than** most other forms of writing.



How Important Is Technical Communication?

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Oral and Written Communication **Skills** are among the **most important** in the technical workplace.

A **survey** in a certain workplace shows that:

- ☐ 40% of employers wanted new hires to **have** better speaking skills;
- ☐ 25% wanted new hires to **have** better writing skills.....